

07-12-2024

Problem with ROS and the Jetson : When calling sleep() within a function, it sometimes happens that the function gets stuck in the sleep() call and never exits. This has something to do with ROS, and we have not found a fix yet. Solution is to restart the jetson.

27-02-2025

Series of issues related to the jetson not turning on. When plugging in the Jetson, the green light turns on and the fan is spinning but there is nothing being displayed when connecting to the screen. The keyboard is not giving any response and attempts at making a connection through serial were unsuccessful.

Further investigation into the problem revealed the following :

- The SD card and the SSD were both corrupted
 - We know this because we attempted to use them in another jetson borrowed from a friend
- The on the jetson itself was also corrupted because putting in a working SD card into the jetson also yielded a black screen
- It is necessary to reflash the Jetpack SDK on the Jetson to be able to make it work again

Flashing the Jetpack SDK onto the Jetson seemed to be the obvious solution to the problem. If we could put a fresh install back onto it, then we should be able to start using it again. This proved to be more complicated for several reasons.

The first difficulty encountered was with respect to getting the Jetson to connect to our computers and to be identified by the SDK. One important step for this to actually work is to short the correct pins : 3 (FC REC) and 4 (GND).

As shown in figure 1, the pins that should be connected are on the side of the board, underneath the CPU and GPU, and are not the upright pins.

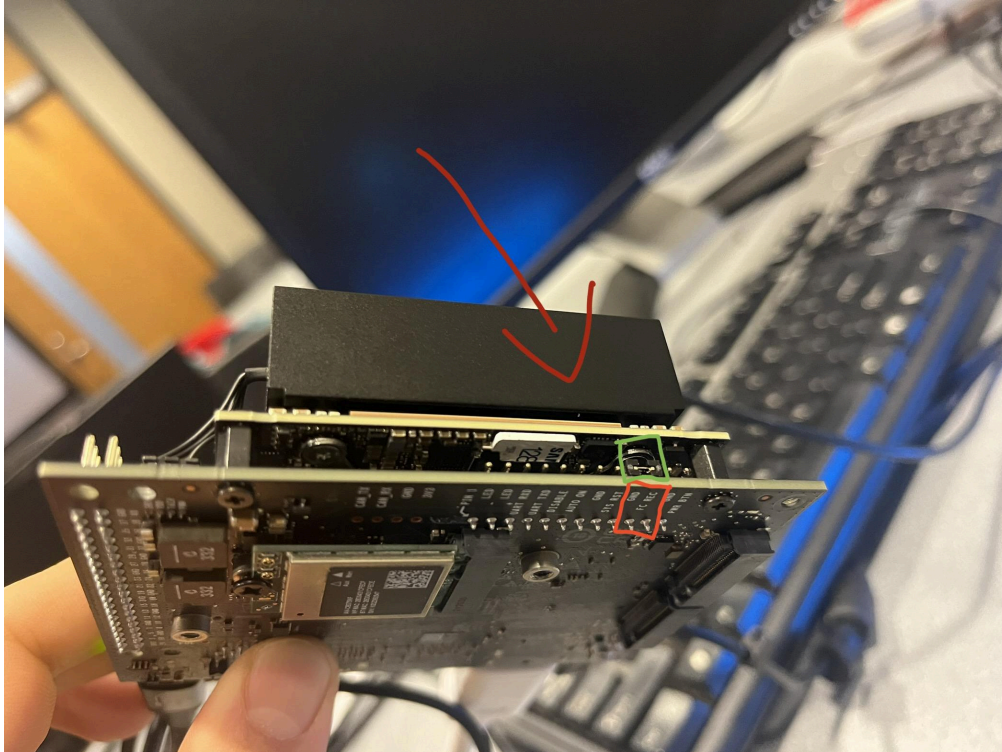


Figure 1 : Pins to be shorted for recovery mode

Once this issue was solved, the next step was to flash the correct version of Jetpack. Given that our algorithm runs on ROS 1, we needed a version of Jetpack that was running Ubuntu 20.04. This meant that we needed to flash an earlier version of Jetpack and not the most recent one. While we had initially flashed the most recent version, just to check if it was still possible to flash onto the Jetson and to get it to work, when we came to attempt to flash the older versions of Jetpack we ran into issues.

Installing Jetpack also ran into the following issue : we were partly successful in installing an earlier version of Jetpack onto the SD card, but unsuccessful when it came to installing it on the nVME SSD.

When attempting to flash using the « old, broken Jetson » (i.e. the Jetson on which the failure happened, and not the Jetson borrowed from Simon's friend), we were able to flash Jetpack 5.1.4 up until Step 3, but we were unable to get the display to work and to therefore to move on to Step 4. Ultimately, we ended up using the « new Jetson » to flash.

This was weird because we were able to flash Jetpack 6.2 on the « old Jetson », and this did seem to work as we got the display working in that way.

A solution may have been found using the following procedure.

Requirements

- Nvidia Jetson Nano
- Monitor for the Jetson Nano's display
- Keyboard and mouse
- USB C-C or C-A cable for connection between computer and Jetson Nano
- Pin jumper
- [Nvidia SDK manager](#)
- 50gb of free space on your computer
- 128gb SD card

Step 1

Short the pins 3 and 4 as shown in figure 1. Power on the Jetson and connect it to your computer.

Also make sure that the device being used to flash the SDK is an SD card and not an nVME SSD. While it would theoretically be possible to flash using the nVME or another device, we have not had success in doing so.

Step 2

On your computer, launch the Nvidia software manager. Your computer should automatically detect that your Jetson Nano board is connected. If not automatically detected, you can check the following :

- In the terminal, use *lsusb* to list all usb devices to see if it is connected
- In the terminal, use *ls /dev/tty** to see all connected devices

If the Nvidia Jetson does not appear when using these commands, then Step 1 was most likely not done properly or your Jetson is obliterated.

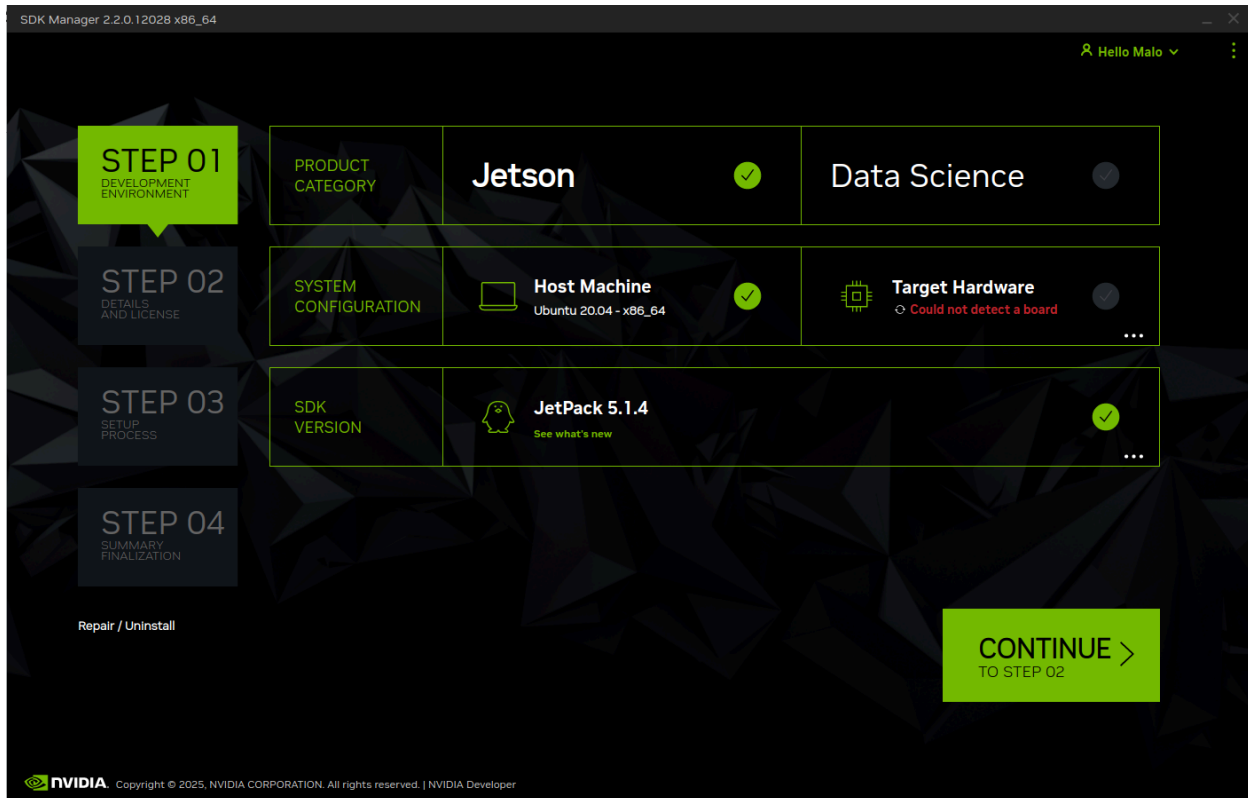


Figure 2 : Computer unable to detect Jetson

```
malo@malo-ThinkPad-X1: ~
malo@malo-ThinkPad-X1: ~ 93x40
malo@malo-ThinkPad-X1:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 04f2:b7e0 Chicony Electronics Co., Ltd Integrated Camera
Bus 003 Device 003: ID 06cb:0123 Synaptics, Inc.
Bus 003 Device 005: ID 8087:0033 Intel Corp.
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 008: ID 0955:7020 NVIDIA Corp. L4T (Linux for Tegra) running on Tegra
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
malo@malo-ThinkPad-X1:~$ ls /dev/tty*
/dev/tty /dev/tty23 /dev/tty39 /dev/tty54 /dev/tty51 /dev/ttyS25
/dev/tty0 /dev/tty24 /dev/tty4 /dev/tty55 /dev/ttyS10 /dev/ttyS26
/dev/tty1 /dev/tty25 /dev/tty40 /dev/tty56 /dev/ttyS11 /dev/ttyS27
/dev/tty10 /dev/tty26 /dev/tty41 /dev/tty57 /dev/ttyS12 /dev/ttyS28
/dev/tty11 /dev/tty27 /dev/tty42 /dev/tty58 /dev/ttyS13 /dev/ttyS29
/dev/tty12 /dev/tty28 /dev/tty43 /dev/tty59 /dev/ttyS14 /dev/ttyS3
/dev/tty13 /dev/tty29 /dev/tty44 /dev/tty6 /dev/ttyS15 /dev/ttyS30
/dev/tty14 /dev/tty3 /dev/tty45 /dev/tty60 /dev/ttyS16 /dev/ttyS31
/dev/tty15 /dev/tty30 /dev/tty46 /dev/tty61 /dev/ttyS17 /dev/ttyS4
/dev/tty16 /dev/tty31 /dev/tty47 /dev/tty62 /dev/ttyS18 /dev/ttyS5
/dev/tty17 /dev/tty32 /dev/tty48 /dev/tty63 /dev/ttyS19 /dev/ttyS6
/dev/tty18 /dev/tty33 /dev/tty49 /dev/tty7 /dev/ttyS2 /dev/ttyS7
/dev/tty19 /dev/tty34 /dev/tty5 /dev/tty8 /dev/ttyS20 /dev/ttyS8
/dev/tty2 /dev/tty35 /dev/tty50 /dev/tty9 /dev/ttyS21 /dev/ttyS9
/dev/tty20 /dev/tty36 /dev/tty51 /dev/ttyACM0 /dev/ttyS22
/dev/tty21 /dev/tty37 /dev/tty52 /dev/ttyprintk /dev/ttyS23
/dev/tty22 /dev/tty38 /dev/tty53 /dev/ttyS0 /dev/ttyS24
malo@malo-ThinkPad-X1:~$
```

Figure 3 : Commands to see if Jetson is properly connected

Step 3

Once the board is connected, choose the Jetpack version you would like to install. Follow the steps that the SDK manager prompts.

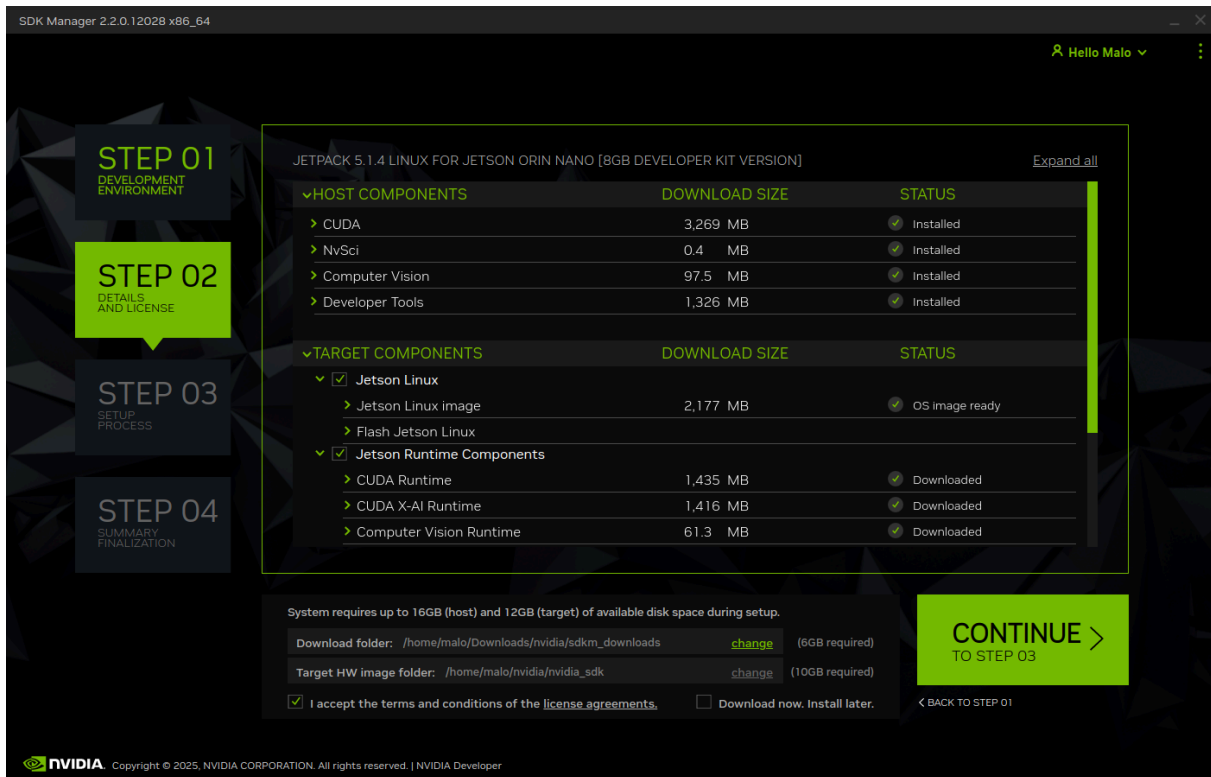


Figure 4 : Getting ready to flash

The flashing takes place in two main stages. The first stage is to install a base Ubuntu operating system on the Jetson. Once that is done, the Jetson should be able to boot and the SDK manager will prompt you with a message.

Step 4

Ubuntu OS having been installed, the message shown in figure 5 should be prompted on your computer. To be able to continue past this without skipping and to install the Jetpack SDK, you should login to the Jetson on the monitor using the keyboard and mouse. Ensure that the correct IP is listed and then attempt the installation.

Note : you should not need to connect to wifi for this. If your computer is connected to the internet, it should be able to provide an internet connection through the USB cable.

Step 5

Once the Jetson has been flashed entirely, disconnect it and power it off. Take out the SD and clone it to the SSD. Then, transfer the SSD to the Jetson and boot it up.

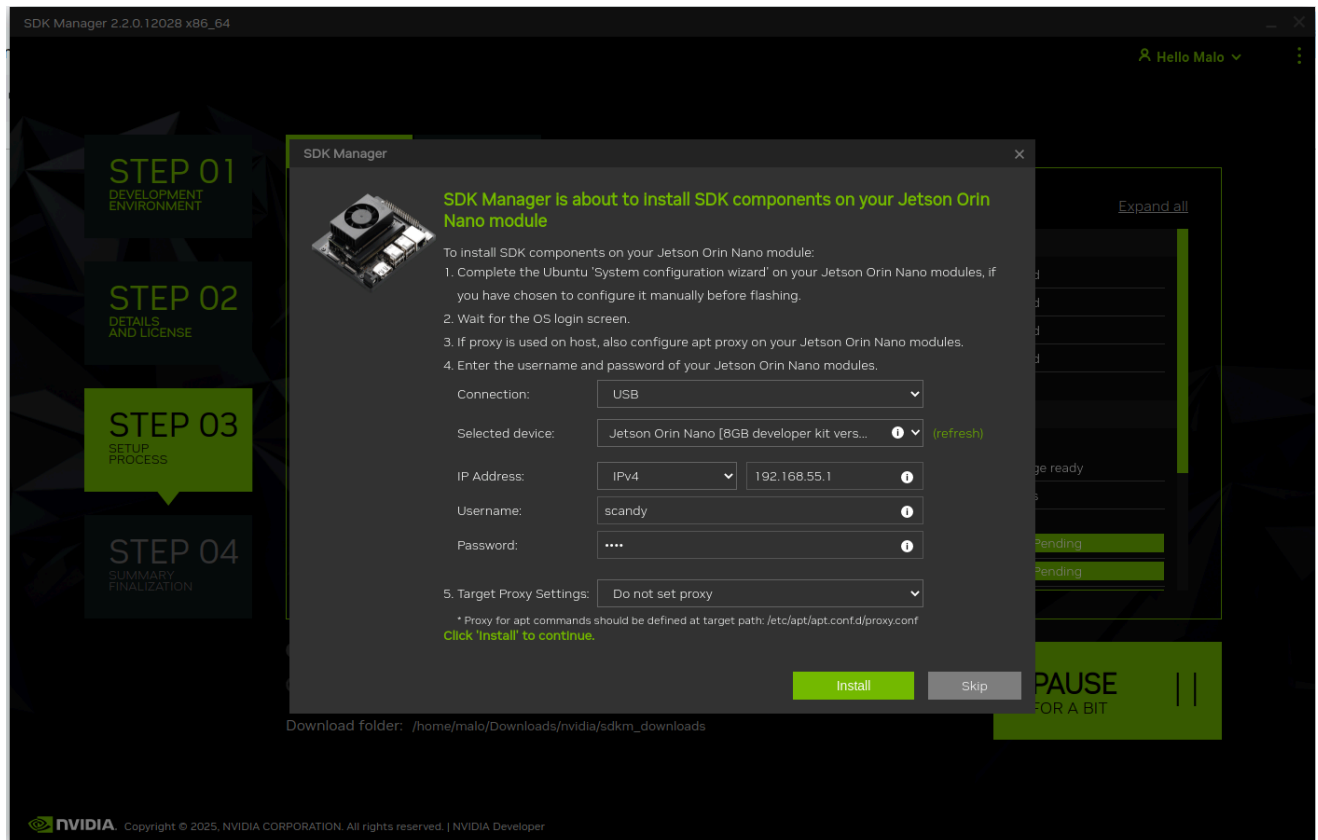


Figure 5 : Prompt to install Jetpack SDK